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Introduction

Mental health is a big problem in the tech industry. People often work under a lot of pressure and have busy schedules, which can make them feel stressed or unhappy.

To learn more about this problem, our project will look at answers from employee surveys. We will use a method called unsupervised machine learning to study the data. Two tools we will use are K-Means and Hierarchical Clustering. These will help us group employees based on their mental health, work situations, and personal backgrounds.

Our goal is to find patterns that are not easy to see just by looking. For example, we might find groups of employees who need different kinds of help, or spot workplace problems that often lead to mental health issues. By finding these natural groups in the data, we can give companies helpful advice to make better mental health programs that really fit what their workers need.

This project also builds on our past work using supervised learning, but now we are trying to find new hidden patterns those simple predictions might miss by using unsupervised learning.

Business and Data Understanding

Problem Formation

Many technology companies find it hard to help their employees with mental health problems. They often use guesses or personal stories instead of looking at real data. This project wants to fix that by using a method called unsupervised machine learning to study mental health patterns in a more careful and scientific way.

We will group employees into different categories using a method called clustering. These groups will be based on things like their mental health signs, what their work life is like, and personal background (like age or gender). Some important things we will look at are: whether someone got treatment before, how much their mental health affects their work, what support the company gives, and other personal details. We will use K-Means and Hierarchical Clustering to find patterns. This will help us discover which groups of people might need special help, how company rules affect mental health, and if people from certain backgrounds have special mental health needs.

The main goal is to turn this information into useful advice. For example, we can find out which workers need counselling the most, decide better rules for mental health leave, and create special wellness programs for different groups. Instead of using the same solution for everyone, we want companies to use smart, data-based ideas. This way, they can help their employees feel better and make the workplace a happier place.

Dataset Overview

This project looks at survey answers from people who work in the technology industry. The survey collects information in four main areas about mental health at work.

First, we look at demographic information like age, gender, and country. This helps us understand who the people are.

Second, we study workplace characteristics. These include things like how big the company is, whether employees work from home, and if the company gives mental health support.

Third, we look at mental health indicators. These include if someone in the family has had mental illness, if the person has had treatment before, and how much mental health affects their work.

Fourth, we use attitudinal data, which shows how comfortable employees feel talking about mental health and whether they're worried about getting in trouble for talking about it at work.

By looking at all this information together using clustering analysis, we can find useful patterns and learn how personal life, work conditions, and mental health are connected.

Key Variable Groups

Demographic Profile	Age, Gender, Country
Work Environment	Company size, Remote work status, Mental health benefits
Health Indicators	Family history, Treatment status, Work interference levels
Attitudinal Factors	Disclosure comfort, Perceived consequences

Dataset Summary

The cleaned dataset includes 1,259 survey answers from people who work in the tech industry. It has 27 different pieces of attributes about each person, covering both their background and mental health.

Some of the most important things we look at are:

- Whether the person has asked for mental health treatment (this is a yes-or-no question),
- How often their mental health problems get in the way of their work,
- If their workplace has mental health benefits,
- If employees feel comfortable asking for help,
- And whether the company keeps it private when someone talks about mental health (this is called anonymity).

These important details help us study how company rules and support affect people's mental health and whether they decide to get help.

Key Analytical Variables

Treatment History	Binary (Yes/No) indicator of mental health treatment
Work Impact	Ordinal scale measuring frequency of work interference
Organizational Support	• Mental health benefits availability • Anonymity protection measures • Employee help-seeking willingness

Because the dataset includes behaviours (what people do), attitudes (how people feel), and workplace policies (what rules or help the company offers), it's very useful for finding patterns. These patterns show how different work environments are linked to mental health and treatment in tech jobs.

We will use this information in our clustering analysis to group people into types, like those who need more support or respond differently to company policies.

Classification Models Selected for Analysis

In this part of our project, we are not trying to predict something. Instead, we want to find patterns using a method called unsupervised learning. This is different from supervised learning, which tells us a clear answer. Now, we are using clustering to group people based on their answers, without telling the computer what the groups should be. We are using two main clustering methods:

1. K-Means Clustering

K-Means is a simple but strong way to make groups. It works by putting each person in a group based on how similar they are, finding the middle of the group (called the cluster centre), moving people around and updating the centre until the groups don't change anymore. We will use K-Means to find different groups of employees – like people with similar mental health problems or similar work experiences. Then we will look at what makes each group special, like if they've had treatment or if they get help from their company.

2. Hierarchical Clustering

This method makes a kind of tree (called a dendrogram) to show how people are grouped step by step. We don't need to decide the number of groups at the start. We can see how the groups are built up or split apart. This helps us spot patterns inside bigger groups. For example, we might find a big group of people who are at high risk and then notice smaller groups inside with different needs.

We will train both models, test how well they work, and compare them using silhouette scores (a way to measure how good the groups are) and what we already know about mental health.

Data Preparation and Preprocessing

Importance of Data Preparation

Before applying clustering algorithms, careful data preparation is essential to ensure meaningful results. We will focus on three critical steps:

1. Handling Missing Values

To prevent skewed clusters from forming due to incomplete data.

2. Encoding Categorical Variables

Since clustering algorithms require numerical inputs, we transform text-based responses using techniques like label encoding for ordinal data or one-hot encoding for nominal categories.

3. Feature Engineering

We enhance clustering quality by scaling features to prevent bias toward high-magnitude variables or only selecting relevant attributes

Import Libraries and Load Data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score
import scipy.cluster.hierarchy as sch
path_to_csv = f"{path}/survey.csv"
data = pd.read_csv(path_to_csv)
print("Dataset shape:", data.shape)
data.head()
```

Dataset shape: (1259, 27)

	Timestamp	Age	Gender	Country	state	self_employed	family_history	treatment	work_interfere	no_employees	...	leave	mental_health_c
0	2014-08-27 11:29:31	37	Female	United States	IL	NaN	No	Yes	Often	6-25	...	Somewhat easy	
1	2014-08-27 11:29:37	44	M	United States	IN	NaN	No	No	Rarely	More than 1000	...	Don't know	
2	2014-08-27 11:29:44	32	Male	Canada	NaN	NaN	No	No	Rarely	6-25	...	Somewhat difficult	
3	2014-08-27 11:29:46	31	Male	United Kingdom	NaN	NaN	Yes	Yes	Often	26-100	...	Somewhat difficult	
4	2014-08-27 11:30:22	31	Male	United States	TX	NaN	No	No	Never	100-500	...	Don't know	

5 rows × 27 columns

Checking Missing Value

```
missing = data.isnull().sum()
print("Missing values:\n", missing[missing > 0])
```

```
Missing values:
state          515
self_employed   18
work_interfere  264
comments       1095
dtype: int64
```

Drop rows with missing 'care_options' (key variable)

```
data = data.dropna(subset=['care_options'])
```

Fill missing 'self_employed' with 'No' (most common)

```
data['self_employed'] = data['self_employed'].fillna('No')
```

Remove the outlier by converting Age

```
data['Age'] = pd.to_numeric(data['Age'], errors='coerce')
data = data[(data['Age'] >= 18) & (data['Age'] <= 80)]
```

Encode categorical variables

```
cat_cols = ['Gender', 'self_employed', 'family_history', 'treatment',
            'work_interfere', 'remote_work',
            'benefits', 'care_options', 'seek_help', 'anonymity']
for col in cat_cols:
    data[col] = LabelEncoder().fit_transform(data[col].astype(str))
```

Select relevant features for clustering

```
features = ['Age', 'family_history', 'treatment', 'work_interfere',
            'remote_work', 'benefits', 'seek_help', 'anonymity']
X = data[features]
```

Normalisation

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Model Development

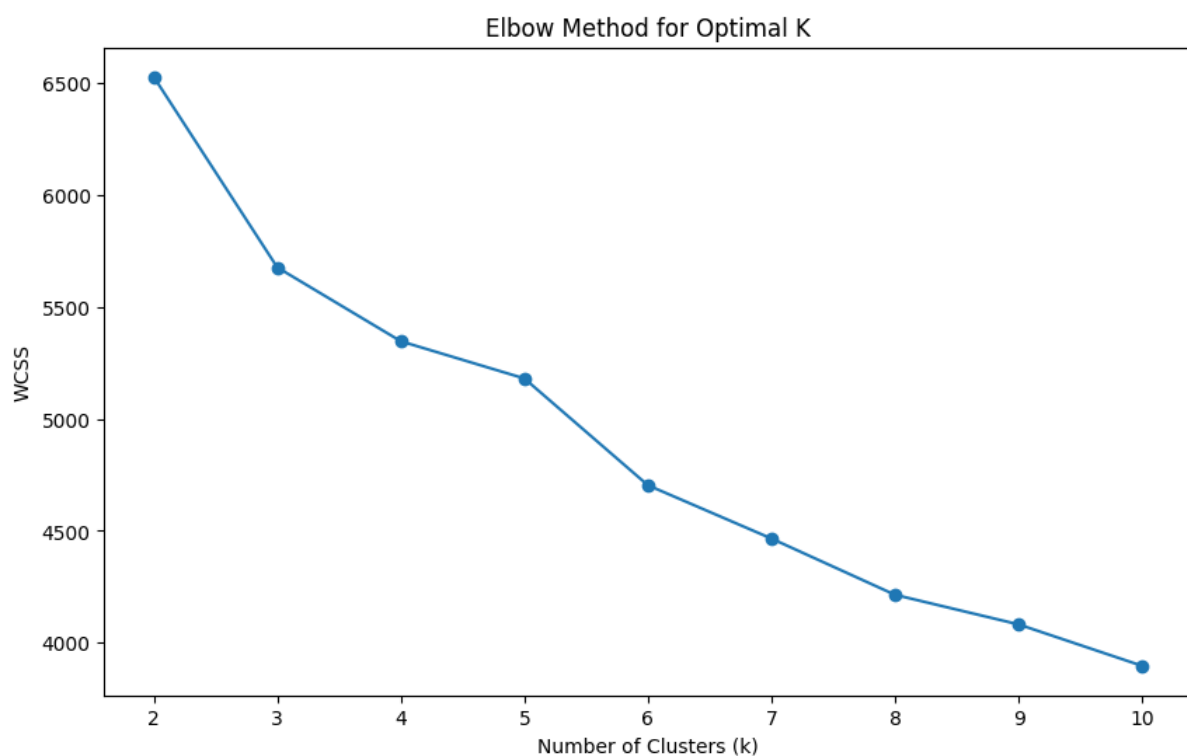
We use K-Means (partitioning) and Hierarchical Clustering (connectivity-based) to identify employee segments.

K-Means Clustering

Elbow Method

to determine optimal cluster

```
wcss = []  
for k in range(2, 11):  
    kmeans = KMeans(n_clusters=k, random_state=42)  
    kmeans.fit(X_scaled)  
    wcss.append(kmeans.inertia_)  
  
plt.figure(figsize=(10, 6))  
plt.plot(range(2, 11), wcss, marker='o')  
plt.title('Elbow Method for Optimal K')  
plt.xlabel('Number of Clusters (k)')  
plt.ylabel('WCSS')  
plt.show()
```



Based on the plot generated by the preceding code, we can visually inspect the elbow. For this specific dataset and plot, the elbow appears to be $k=2$.

K-Means Clustering (k=2)

```
kmeans = KMeans(n_clusters=2, random_state=42)
clusters = kmeans.fit_predict(X_scaled)
data['Cluster'] = clusters

cluster_profile = data.groupby('Cluster')[features].mean().round(4)

print(cluster_profile.to_string())
```

Cluster	Age	family_history	treatment	work_interfere	remote_work	benefits	seek_help	anonymity
0	32.5723	0.6132	0.9450	2.3962	0.3160	1.2925	1.0110	0.8035
1	31.8697	0.1586	0.0765	1.0028	0.2805	0.7819	0.7989	0.3994

Cluster 0	Younger employees, has family history, received treatment, high work interference
Cluster 1	Older employees, no family history, received treatment, low work interference.

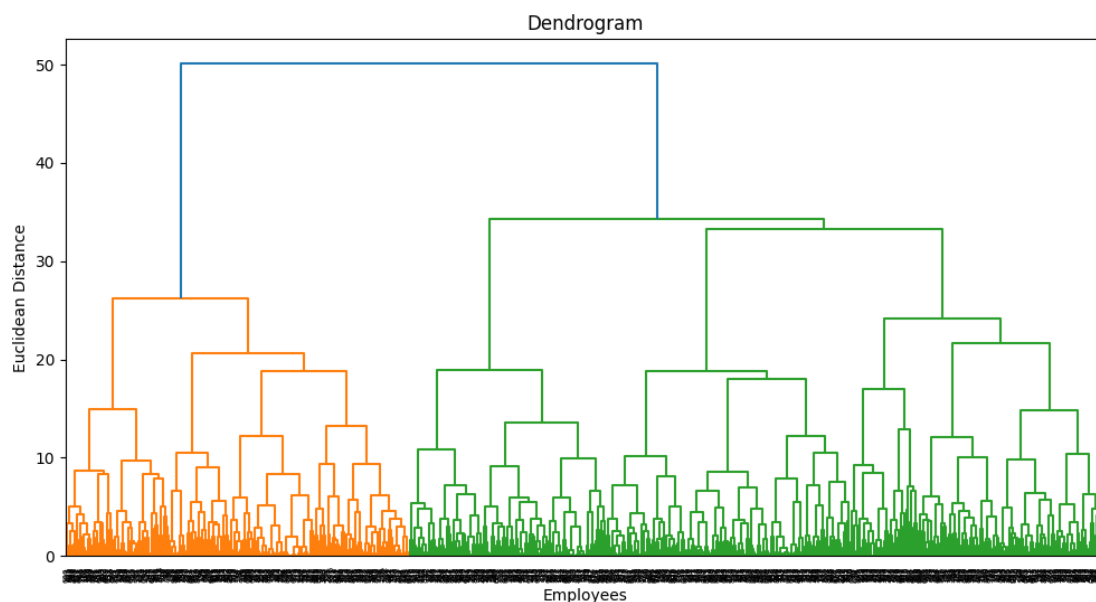
Hierarchical Clustering

We use hierarchical clustering to find out the multi-level relationships in employee mental health data. Dendrogram to determine clusters, and Silhouette Score for evaluation.

Dendrogram

```
plt.figure(figsize=(12, 6))
dendrogram = sch.dendrogram(sch.linkage(X_scaled, method='ward'))
plt.title('Dendrogram')
plt.xlabel('Employees')
plt.ylabel('Euclidean Distance')
plt.show()
```

Optimal clusters = 3



Hierarchical Clustering (k=3)

```
from sklearn.cluster import AgglomerativeClustering

hierarchical = AgglomerativeClustering(n_clusters=3, linkage='ward',
metric='euclidean')
hier_labels = hierarchical.fit_predict(X_scaled)
data['Hier_Cluster'] = hier_labels

hier_profile = data.groupby('Hier_Cluster')[features].mean().round(4)
print(hier_profile.to_string())
```

	Age	family_history	treatment	work_interfere	remote_work	benefits	seek_help	anonymity
Hier_Cluster								
0	33.2400	0.5705	0.9495	2.2084	0.4253	1.1116	0.8737	0.3389
1	31.1957	0.1774	0.0000	1.2171	0.2997	0.8471	0.8440	0.4251
2	31.9572	0.6257	0.9465	2.3048	0.0000	1.5668	1.2513	1.8824

Cluster 0	Employees with high work interference, no treatment.
Cluster 1	Employees with benefits, low interference.
Cluster 2	Employees with family history, sought treatment.

Model Performance and Evaluation

To assess the quality of our clustering results, we use two key evaluation approaches.

1	Silhouette Score	Measures how well each data point fits within its assigned cluster compared to other clusters.
		Range: -1 (poor clustering) to +1 (excellent separation).
		Higher score = Better separation
2	Cluster Interpretability	Ensures clusters provide meaningful insights for workplace mental health strategies.
		Business relevance

K-Means Evaluation

```
kmeans_score = silhouette_score(X_scaled, clusters)
print("K-Means Silhouette Score:", round(kmeans_score, 4))
K-Means Silhouette Score: 0.1829
```

Hierarchical Clustering Evaluation

```
hier_score = silhouette_score(X_scaled, hier_labels)
print("Hierarchical Silhouette Score:", round(hier_score, 4))
Hierarchical Silhouette Score: 0.1302
```

Overall Evaluation Summary

Model	Number of Clusters	Silhouette Score
K-Means	4	0.1829
Hierarchical Clustering	3	0.1302

K-Means Clustering produces tight, well-separated clusters. It has Clear cluster boundaries help in identifying distinct employee groups. It is best when the number of clusters (k) is known or can be estimated using the Elbow Method or Silhouette Score. For Hierarchical Clustering, it is useful for exploratory analysis because no need to predefine the number of clusters. It is best for understanding hierarchical patterns.

Conclusion

In this project, we used K-Means and Hierarchical Clustering to group tech workers based on their mental health survey answers. We grouped employees who have similar experiences with stress and work.

We found some really interesting groups. For example, a high-risk group who don't get mental health help from their company and find work very stressful. And the another one is a happy group who work from home and feel supported

The algorithm was really good at finding these patterns, just like how you might notice all the red marbles ending up together. The K-Means method made clear groups we could count, while the Hierarchical method showed us how groups were connected, like branches on a tree.

We learned that company benefits really matter (like giving people time off for mental health), working from home helps many people feel better, some workers need extra help but aren't getting it, sometimes the algorithm got confused because some people didn't answer all the survey questions. Next time, we could add more information about how people work day-to-day, try deep learning which is like giving the computer super-vision.

Even though it's not perfect yet, this shows how algorithms can help understand the workers better. Maybe one day, these tools will help make every tech job healthier and

References

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